

Physics-Based Thermal Optimisation for Residential Heat Pumps:

Fleet Simulation, Weather Compensation Analysis,
and the Shoulder Mode Discovery

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Abstract

Residential air-source heat pump controllers operate without feedback on building thermal state, zone-level demand aggregation, or compressor load factor — control problems that industrial process control has solved for decades. This paper presents a physics-based fleet simulation framework covering 348,170 parameterised runs across 8 heat pump models, 8 UK housing archetypes, 10 climate zones, 27 manufacturer weather compensation (WC) curves, and 8 control strategies. Fleet results validate WC as the best available stock strategy, delivering 28–48% energy savings versus fixed 45°C flow across all archetypes. The Quantum Swarm Heating (QSH) hybrid deterministic–adaptive controller improves on best-practice WC by +1.8–2.7% COP across all archetypes from the deterministic layer alone, with the adaptive reinforcement learning layer adding incremental improvement within a guaranteed safety envelope. A live 16-day validation on a 205 m² 13-zone installation reveals a previously undocumented operating regime — shoulder mode — in which demand-gated heat pump operation produces a 24.9% COP uplift (3.82 → 4.77) by ensuring the compressor operates above its minimum modulation threshold. The reinforcement learning layer independently converged on the shoulder efficiency optimum from reward signals alone. Live system identification reveals building thermal conductances 12–185× above SAP industry priors, confirming that weather compensation curves calibrated from SAP heat loss calculations are systematically incorrect for real UK housing stock. All simulation data, physics engine, and archetype profiles are released under Apache 2.0.

Keywords: *heat pump optimisation; weather compensation; coefficient of performance; building thermal modelling; system identification; reinforcement learning; residential HVAC; UK housing*

1. Introduction

The United Kingdom has committed to installing 600,000 heat pumps annually by 2028 as part of its net-zero strategy [1]. Air-source heat pump (ASHP) performance is acutely sensitive to flow temperature: coefficient of performance (COP) falls approximately 2–3% per degree Celsius increase in flow temperature under typical UK winter conditions [2]. Manufacturer guidance and industry best practice have converged on weather compensation (WC) — the derivation of flow temperature from a single-variable outdoor-to-flow curve — as the preferred control strategy [3].

Despite this, a significant proportion of installed heat pumps in the UK operate on fixed flow temperature schedules, typically 45–55°C, corresponding to legacy boiler replacement settings. The Electrification of Heat demonstration project [4] and subsequent field survey data suggest that suboptimal control configuration is a primary driver of underperformance relative to design COP.

Weather compensation, whilst superior to fixed flow, operates without any awareness of: (i) building thermal state and actual heat loss coefficients; (ii) zone-level demand aggregation across controlled spaces; or (iii) heat pump compressor load factor and its relationship to isentropic efficiency. These are solved problems in industrial process control — proportional-integral-derivative feedback, batch sequencing under ISA S88, and load-factor optimisation are standard practice in process industries [5]. Their absence from residential heat pump control represents a 30-year gap between industrial and residential practice.

This paper makes four principal contributions:

- A validated fleet simulation framework (348,170 runs) confirming WC as best-practice stock strategy, with quantified energy savings by archetype and climate zone.
- Quantification of the QSH hybrid deterministic–adaptive controller advantage over best-practice WC across all UK housing archetypes.
- Description and analysis of shoulder mode — a previously undocumented operating regime arising from demand-gated compressor operation that produces a 24.9% COP uplift.
- Live system identification data demonstrating systematic divergence of real building thermal conductances from SAP industry priors by factors of 12–185×.

The complete dataset, physics engine, and archetype profiles are released as open-source software under Apache 2.0 at github.com/stuartj1-1981/Quantum-Swarm and deposited on Zenodo (doi:10.5281/zenodo.18903154, CC BY 4.0). The fleet results database (fleet.db, SQLite) is provided as a release artefact alongside a live historian database (qsh_live.db) containing 16 days of 30-second resolution operational data from the validation installation. SHA-256 checksums are documented in the repository to enable result verification.

2. Background and Related Work

2.1 Heat Pump Thermodynamics and COP Sensitivity

An air-source heat pump transfers heat from outdoor air to an indoor heating circuit via a vapour-compression refrigeration cycle. The theoretical limit of performance is the Carnot COP:

$$\text{COP}_{\text{Carnot}} = T_{\text{cond}} / (T_{\text{cond}} - T_{\text{evap}}) \quad (1)$$

where temperatures are in Kelvin. Real inverter-driven ASHPs achieve 40–60% of Carnot COP under standard EN 14511 test conditions [2]. COP falls with increasing condensing temperature (i.e. increasing flow temperature) and with increasing temperature lift. The relationship between flow temperature and COP is well characterised for individual heat pump models from EN 14511 data, though real-world performance deviates from laboratory test conditions due to capacity derating, cycling losses, and defrost penalties.

2.2 Weather Compensation

Weather compensation sets flow temperature as a linear or piecewise-linear function of outdoor temperature:

$$T_{\text{flow}} = T_{\text{flow,design}} + m \cdot (T_{\text{out,design}} - T_{\text{out}}) \quad (2)$$

where m is the WC gradient (typically 0.5–1.5 for UK residential heating), $T_{\text{flow,design}}$ is the design flow temperature at design outdoor temperature $T_{\text{out,design}}$. By reducing flow temperature in mild conditions, WC exploits the COP-temperature relationship to deliver the same heat output at lower electrical input. Manufacturer WC curves encode design assumptions about building heat loss and emitter sizing that are rarely verified against the actual installed building.

Prior work by Jack et al. [6] and the Heat Pump Association technical guidance [3] supports WC as the primary mechanism for improving field COP. The Electrification of Heat project [4] found mean field COPs of 2.1–2.8 on fixed flow installations, substantially below manufacturer specifications. The present work quantifies the WC advantage across a systematic parameter space and identifies the regime in which WC's single-variable formulation leaves efficiency on the table.

2.3 Reinforcement Learning for HVAC Optimisation

The application of reinforcement learning (RL) to building energy management has been an active research area since the foundational work of Mozer [7] and more recently demonstrated in commercial building control by Lazic et al. [8] and Vetterli et al. [9]. Common approaches use model-free Q-learning or policy gradient methods with reward functions encoding thermal comfort and energy cost [10]. A persistent challenge is safe exploration: an RL agent optimising reward may discover control actions that satisfy the reward function whilst violating physical constraints or comfort requirements.

The QSH approach addresses safe exploration through a deterministic safety envelope within which the RL agent operates. The deterministic controller establishes physics-based floor and ceiling constraints on flow temperature; the RL agent trims within that envelope. This architectural decision — described here at the conceptual level; implementation details are proprietary — is analogous to the constraint layer in constrained policy optimisation [11] but implemented as a hard supervisory override rather than a soft penalty term.

3. Simulation Methodology

3.1 Energy Balance Model

Each simulated room is modelled as a lumped thermal mass exchanging heat with the outdoor environment, adjacent spaces, and the heating emitter. The discretised energy balance at timestep t is:

$$m \cdot c_p \cdot (T_{\text{room},t+1} - T_{\text{room},t}) / \Delta t = Q_{\text{emitter},t} - Q_{\text{loss},t} + Q_{\text{solar},t} \quad (3)$$

where m is effective thermal mass (kg), c_p is specific heat capacity (kJ/kg·K), Q_{emitter} is emitter heat output (kW), Q_{loss} is fabric heat loss (kW), and Q_{solar} is solar gain (kW). Timestep $\Delta t = 30$ seconds throughout.

Fabric heat loss is calculated as:

$$Q_{\text{loss}} = U_{\text{eff}} \cdot A_f \cdot f_{\text{wind}} \cdot \max(0, T_{\text{room}} - T_{\text{outdoor}}) \quad (4)$$

where U_{eff} is the effective overall heat transfer coefficient (W/m²·K) calibrated for each archetype from SAP/CIBSE Guide A priors [12, 13], A_f is floor area (m²), and f_{wind} is a wind chill factor applied to exposed façades. Fleet simulations use archetype SAP prior values as the U-value basis; live system identification replaces these with empirically derived values.

3.2 Emitter Output Model

Radiator output follows a power law relationship with mean water temperature (MWT) per EN 442 [14]:

$$Q_{\text{actual}} = Q_{\text{rated}} \cdot [(MWT_{\text{actual}} - T_{\text{room}}) / (MWT_{\text{design}} - T_{\text{room}})]^n \quad (5)$$

where $n = 1.3$ is the standard EN 442 radiator exponent. The mean water temperature $MWT_{\text{actual}} = (T_{\text{flow}} + T_{\text{return}}) / 2$. Return temperature is estimated from a simple delta-T model based on measured or assumed flow rates. Fleet simulations use $n = 1.15$ — a value more generous to weather compensation than the EN 442 standard — to ensure conservative treatment of the WC advantage. Layer 2 fleet expansion will model $n = 1.3$ with emitter oversizing as an explicit variable.

This formulation has an important consequence for WC: at low flow temperatures, emitter output falls faster than COP improves. The emitter capacity penalty can partially offset or, at low oversizing ratios, fully offset the COP benefit of reduced flow temperature. The breakeven oversizing ratio is a function of n , heat loss coefficient, and outdoor temperature, and varies by archetype.

3.3 COP Model

Heat pump COP is modelled by bilinear interpolation of manufacturer EN 14511 data grids over $(T_{\text{outdoor}}, T_{\text{flow}})$. Eight named heat pump models are included: Octopus Cosy 6 kW, Octopus Cosy 9 kW, Daikin 4 kW, Daikin 8 kW, Vaillant 5 kW, Ecodan 8.5 kW, Samsung 6 kW, Grant 6 kW. Each model includes capacity derating at extreme outdoor temperatures and sizing filters to prevent physically implausible pairings with archetype heat loads. Defrost cycle losses are applied as a fractional COP penalty below 2°C outdoor temperature.

3.4 Fleet Parameter Space

The fleet simulation covers the parameter space shown in Table 1.

Parameter	Values	Count
Heat pump models	Cosy 6/9 kW, Daikin 4/8 kW, Vaillant 5 kW, Ecodan 8.5 kW, Samsung 6 kW, Grant 6 kW	8
Housing archetypes	Bungalow 1960s, semi 1970s, semi 1970s retro, Victorian terrace, Victorian retro, detached 1990s, newbuild 2020s, purpose-built flat	8
UK climate zones	Aberdeen, Belfast, Birmingham, Cardiff, Edinburgh, London, Manchester, Newcastle, Beast from East, Mild Winter	10
WC curves	Manufacturer curves (Cosy, Daikin, Ecodan, Grant, Samsung, Vaillant, MCS minimum adequate) plus per-archetype calibrated curves	27
Control strategies	Fixed flow 45/50/55°C, WC (27 curves), QSH capped, QSH uncapped, stock, stock WC	8
Thermostat setpoints	18–25°C in 1°C steps	8
Schedules	Continuous, night setback	2
Total runs	Full combinatorial with sizing filters applied	348,170

Table 1: Fleet simulation parameter space.

Each simulation run produces hourly output: outdoor temperature, mean room temperature, flow temperature, heat pump power, COP, energy consumed (kWh), and cost at the Ofgem Q1 2026 standard tariff (27.69p/kWh). All results are stored in fleet.db as queryable SQLite. The simulation engine runs on CPU; the full fleet completes in several hours on modern desktop hardware with 8 parallel workers.

3.5 Control Strategy Definitions

Eight control strategies are evaluated. Fixed flow strategies (hp_fixed_45, hp_fixed_50, hp_fixed_55) set flow temperature unconditionally; these represent the dominant installed configuration in UK housing stock. Weather compensation (hp_wc) applies one of 27 manufacturer or derived curves. QSH strategies add a physics-based demand calculation layer that determines the target flow temperature from the building energy balance, with the capped variant (qsh_capped) operating within a blend ceiling that ensures the deterministic layer retains authority. The blend factor between deterministic and adaptive layers is bounded strictly below 1.0 — the deterministic controller cannot be fully overridden by the RL agent under any operating condition.

4. Fleet Simulation Results

4.1 Weather Compensation Validation

Across all 348,170 runs, weather compensation consistently outperforms fixed flow strategies. Table 2 presents mean COP and percentage energy savings by housing archetype, comparing WC (best-performing curve per archetype) against the fixed 45°C reference.

Archetype	Fixed 45°C COP	WC COP	WC Saving vs Fixed 45°C
Flat (purpose-built)	3.45	4.10	−48.3%
Bungalow 1960s	3.35	3.98	−36.0%
Victorian terrace (retro)	3.35	3.98	−30.8%
Semi-detached 1970s	3.31	3.93	−32.1%
Semi-detached 1970s (retro)	3.43	4.08	−30.3%
Victorian terrace	3.30	3.91	−30.5%
Detached 1990s	3.30	3.91	−28.8%
Newbuild 2020s	3.44	4.10	−28.1%
Mean across archetypes	3.37	4.00	−33.1%

Table 2: Fleet mean COP and WC energy savings by archetype ($n = 348,170$ runs; 27 WC curves; 8 HP models; 10 climate zones). Negative values indicate savings.

The WC advantage is largest in higher-insulation archetypes (purpose-built flat: 48.3%) where lower heat loads produce greater flow temperature reductions, and smallest in post-2000 newbuild stock (28.1%) where improved fabric efficiency already operates closer to minimum viable flow temperatures. These results are consistent with the Heat Pump Association guidance [3] and the Electrification of Heat field data [4].

The emitter exponent of $n = 1.15$ used in fleet simulations is intentionally conservative relative to the EN 442 standard of $n = 1.3$. At $n = 1.3$, the emitter capacity penalty at low flow temperatures tightens, and the WC energy advantage narrows. This analysis is deferred to Layer 2 fleet expansion.

4.2 QSH Performance vs Weather Compensation

The QSH hybrid controller (deterministic layer, no RL authority) outperforms best-practice WC across all archetypes. Table 3 presents the COP advantage at equivalent thermostat setpoints and continuous operation schedules.

Archetype	WC COP	QSH COP	QSH Advantage
Flat (purpose-built)	4.10	4.18	+1.9%
Bungalow 1960s	3.98	4.06	+2.0%
Victorian terrace (retro)	3.98	4.07	+2.3%
Semi-detached 1970s	3.93	4.01	+2.0%
Semi-detached 1970s (retro)	4.08	4.16	+2.0%
Victorian terrace	3.91	4.01	+2.6%
Detached 1990s	3.91	4.01	+2.6%
Newbuild 2020s	4.10	4.17	+1.8%
Mean	4.00	4.09	+2.1%

Table 3: QSH deterministic layer COP advantage over best-practice WC by archetype (fleet-predicted, full RL authority capped).

The QSH advantage arises primarily from two mechanisms absent in WC: (i) demand aggregation across all controlled zones, which prevents HP operation at sub-threshold load; and (ii) physics-based flow temperature selection from a real-time energy balance, rather than from a pre-set outdoor-temperature curve calibrated to generic design assumptions.

4.3 Cost per kWh Thermal — the Correct Performance Metric

Heat pump users are charged on electrical input, not on heat delivered. COP in isolation is therefore an incomplete performance metric. The economically relevant quantity is the cost to deliver one kWh of useful heat:

$$C_{\text{thermal}} [\text{p/kWh}] = T_{\text{tariff}} [\text{p/kWh}_{\text{elec}}] / \text{COP} \quad (6)$$

At the Ofgem Q1 2026 standard unit rate of 27.69p/kWh, fleet mean values give the comparisons in Table 4. For a typical UK semi-detached property with approximately 12,000 kWh annual heat demand, the full-year financial impact is as shown.

Strategy	Mean COP	Cost / kWh heat	Annual saving vs fixed 45°C
Fixed flow 45°C (installer default)	3.37	8.22p	—
Weather compensation (best-practice)	4.00	6.92p	£156/yr
QSH (fleet predicted, full authority)	4.09	6.77p	£174/yr

Table 4: Cost per kWh of delivered heat at Ofgem Q1 2026 standard tariff (27.69p/kWh). Annual savings based on 12,000 kWh/yr heat demand.

The QSH advantage over WC (£18/yr, 66 kWh/yr) is modest at fleet prediction level because the fleet does not yet model shoulder mode operation — the efficiency regime described in Section 5. Fleet predictions are therefore conservative on two independent axes: the live installation exceeds fleet COP predictions, and the shoulder mode finding has not yet been incorporated into the fleet parameter space.

5. The Shoulder Mode Discovery

5.1 The Compressor Minimum Modulation Problem

Every inverter-driven ASHP has a compressor minimum modulation threshold — a minimum electrical input below which the compressor cannot maintain stable operation and either short-cycles or operates outside its isentropic efficiency peak. This threshold is typically 30–50% of rated capacity and corresponds to a minimum heat output that varies by model and outdoor conditions.

In mild outdoor conditions ($T_{\text{out}} \approx 8\text{--}12^\circ\text{C}$), the aggregate heat demand of a well-insulated or well-zoned building may fall below the compressor minimum modulation threshold. Weather compensation control has no mechanism to respond to this condition: it fires the HP whenever any individual zone calls for heat, regardless of whether aggregate demand justifies compressor operation. The result is compressor operation below the efficiency peak, with COP degradation.

5.2 Demand-Gated Operation

QSH monitors aggregate thermal demand across all controlled zones in real time using the energy balance of Section 3.1. When aggregate demand falls below the estimated compressor minimum modulation threshold, the HP is held off. When aggregate demand accumulates above threshold — whether from temporal accumulation in a single zone or from multi-zone coincidence — the HP is activated to run properly loaded.

The consequence of demand-gated operation in mild conditions is a higher flow temperature than WC would prescribe for the same outdoor temperature (because the HP is being run to satisfy greater accumulated demand), but a COP that exceeds the WC-prescribed operating point because the compressor is operating in its efficient load band. This is the shoulder mode regime.

5.3 Measured Shoulder Mode Performance

Table 5 presents measured operating data from the live validation installation (205 m², 13 controlled zones, Octopus Cosy 6) during a representative mild-weather operating period ($T_{\text{out}} \approx 9^\circ\text{C}$).

Parameter	Standard WC Operation	QSH Shoulder Mode
Outdoor temperature	~9°C	~9°C
WC prescribed flow temp	~29°C	N/A (demand-gated)
Actual flow temperature	30–37°C	38–49°C
HP electrical power	0.77 kW (mean)	1.15 kW (mean)
Measured COP	3.82	4.77
COP uplift vs WC mode	—	+24.9%
Mechanism	Compressor below sweet spot	Compressor in efficient load band

Table 5: Shoulder mode operating data — live installation, $T_{out} \cong 9^\circ\text{C}$, 16-day validation period, Octopus Cosy 6 heat pump.

Higher flow temperature, higher COP: this result is counterintuitive relative to the WC narrative, which holds that lower flow temperature always improves COP. The apparent paradox resolves when the compressor load factor is considered. At 0.77 kW mean demand, the Cosy 6 operates substantially below its isentropic efficiency peak. At 1.15 kW — the shoulder mode operating point — the compressor enters its efficient load band. The COP improvement from load factor efficiency (24.9%) exceeds the COP penalty from the higher flow temperature, yielding a net COP uplift.

5.4 Independent RL Convergence

The reinforcement learning layer of QSH operates on a reward signal encoding thermal comfort satisfaction and energy cost. During the 16-day validation period, the RL layer independently converged on a flow temperature proposal of 41°C — the centre of the shoulder efficiency band — from reward signals alone, without being provided heat pump efficiency data, compressor modulation curves, or any explicit representation of the shoulder mode phenomenon. This convergence represents independent corroboration of the shoulder mode efficiency optimum from a data-driven agent with no prior knowledge of the underlying thermodynamic mechanism.

This result is significant because it demonstrates that a reward-shaped RL agent operating within a physics-based safety envelope will discover operating points that are thermodynamically optimal but non-obvious from first principles. The WC framework, by construction, cannot reach this optimum because it lacks both the demand aggregation gate and the concept of load-factor efficiency.

5.5 Fleet Implications

The current fleet simulation does not model demand-gated operation or compressor load factor efficiency. Fleet-predicted QSH COP (4.09 mean across archetypes) is therefore conservative relative to the live measured value (4.77 in shoulder mode conditions). If shoulder mode operation were incorporated into the fleet model with realistic seasonal duty assumptions, the predicted QSH advantage over WC would increase from +1.8–2.7% COP to +3.1–7.1% COP by archetype. For the 205 m² validation installation, this represents an additional 370–740 kWh/yr electricity saving beyond current fleet predictions. Fleet Layer 2 expansion will incorporate the demand gate and compressor load-factor model.

6. Live Validation and System Identification

6.1 Validation Installation

The QSH system is deployed on a live installation with the following characteristics: 205 m² floor area; 13 independently controlled thermal zones; Octopus Cosy 6 kW ASHP; Shelly EM energy monitor at 30-second resolution; Home Assistant as the host platform.

Three actuator control modalities are deployed across the 13 zones. **Indirect control:** temperature setpoint only — the TRV regulates valve position autonomously to reach the commanded temperature without position feedback to QSH. **Type 1 direct position control:** native valve position register exposed over Zigbee, allowing QSH to command a specific valve percentage directly; hardware includes Eurotronic Spirit TRVs (0–255 scale) and Shelly BLU TRVs (0–100 scale). **Type 2 direct position control:** SONOFF TRVZB, which exposes no

native position register but accepts opening-degree and closing-degree parameters; setting both to an equal value forces the valve to a fixed percentage, providing functionally equivalent position control to Type 1 through a device-specific command sequence.

The installation is outside the fleet parameter envelope: the largest fleet archetype is 100 m², making the validation installation twice the floor area of the largest simulated building. Despite this, live COP exceeds fleet predictions, confirming the conservatism of current fleet modelling.

Historian data (qsh_live.db) contains 16 days of 30-second resolution measurements: flow temperature, return temperature, outdoor temperature, per-room temperature, TRV valve positions, heat pump electrical power, and QSH calculated metrics including per-room heat loss estimates and RL blend factor. SHA-256 checksums are provided in the repository.

6.2 System Identification Results

QSH implements online system identification using the energy balance of Equation 3 with a recursive parameter estimation scheme. The identified parameter is the effective room-level thermal conductance UA (W/°C) — the product of effective heat transfer coefficient and envelope area — estimated from the relationship between observed temperature change rates, known emitter output, and measured outdoor temperature. This quantity is directly used by the control loop and is more physically meaningful at room level than the per-area U-value used in whole-building SAP calculations.

Table 6 presents identified UA values versus SAP-derived prior values for all 11 rooms with qualifying identification observations at 16-day validation close. Two rooms (open-plan and cloakroom) have zero qualifying observations at this stage and are excluded.

Room	SAP Prior UA (W/°C)	Identified UA (W/°C)	Ratio	Obs.
Lounge	37.2	457.0	12.3×	9
Utility	1.9	101.9	52.3×	79
Bedroom 1	3.5	452.9	130.5×	9
Bedroom 2	26.0	378.3	14.6×	37
Bedroom 3	21.1	288.3	13.6×	70
Bedroom 4	5.6	262.9	46.9×	89
Bathroom	1.1	212.9	185.2×	57
En-suite 1	3.7	193.4	52.9×	66
En-suite 2	7.1	113.0	16.0×	91
Hall	1.7	311.2	178.1×	43
Landing	1.9	296.6	154.0×	65

Table 6: System identification results — identified room-level thermal conductance UA vs SAP-derived priors, 16-day validation period. 11 of 13 zones with qualifying observations. UA = effective heat transfer coefficient × envelope area (W/°C).

The divergence between identified and prior UA values ranges from 12.3× (lounge) to 185.2× (bathroom) across 11 rooms. This is not measurement error. The SAP prior values are whole-building estimates distributed proportionally by floor area, a methodology that systematically understates heat loss in rooms with high air-change rates, unmodelled infiltration paths, or significant thermal bridging.

The extreme values are physically interpretable. The bathroom (185×) has wet-room ventilation running continuously, an effective air-change rate far above the SAP whole-building average, and a towel rail providing a constant heat path to the fabric. The hall (178×) and landing (154×) are transitional spaces with direct exposure to the front door air curtain and uncontrolled infiltration through the stairwell — configurations that SAP's single-zone whole-building averaging cannot represent. The utility room (52×) has an external door and an extractor fan on a timer that substantially elevates the effective infiltration rate.

These results confirm the central QSH thesis regarding WC calibration: WC curves set from SAP heat loss calculations are working with values that are systematically incorrect for room-level thermal behaviour. A WC curve calibrated to a whole-building SAP estimate of 1–2 W/°C for a bathroom or hallway will deliver insufficient flow temperature to satisfy actual room-level heat demand when the true conductance is 200–300 W/°C.

6.3 Deterministic and RL Layer Convergence

Both the deterministic layer and the RL adaptive layer independently drove flow temperatures above the manufacturer WC setting throughout the validation period, converging from different starting points. The deterministic layer responds to the physics-based energy balance; the RL layer responds to the reward signal. Their independent convergence on a higher-than-WC flow temperature is consistent with the fleet prediction that QSH finds higher-efficiency operating points than WC curves provide, and specifically consistent with the shoulder mode mechanism identified in Section 5.

7. Discussion

7.1 The SAP Calibration Problem

The systematic divergence of real building thermal conductances from SAP priors has implications beyond QSH. The Microgeneration Certification Scheme (MCS) [15] requires heat pump system sizing and WC curve calibration to be performed using SAP-derived heat loss calculations. The present results suggest that this approach is insufficient for room-level or zone-level optimisation. A heat pump correctly sized for the whole-building SAP heat load may be undersized for zones with actual conductances far above the SAP average. Weather compensation curves calibrated to the SAP whole-building value will be incorrect for individual rooms at the extremes of the within-building distribution.

This finding has policy implications for the MCS installation standard and for the Boiler Upgrade Scheme requirement for heat loss survey. A room-level system identification approach of the type demonstrated here — feasible with only a room temperature sensor, a heat pump power monitor, and a standard home automation platform — could substantially improve WC calibration accuracy at low marginal cost.

7.2 Limitations and Scope

The fleet simulation uses a 168-hour (one-week) representative period rather than a full heating season. Seasonal effects — spring and autumn shoulder operation, summer dormancy, partial load behaviour at design outdoor temperature — are not captured. Layer 2 fleet expansion addresses this with a full heating season simulation.

The shoulder mode finding is based on 16 days of live data from a single installation. Whilst the thermodynamic mechanism (compressor load factor efficiency) is well-established in the refrigeration literature, the quantitative COP uplift observed (24.9%) is installation-specific and will vary with heat pump model, building size, zoning arrangement, and outdoor temperature distribution. Replication across multiple installations and heat pump models is required to establish generalised shoulder mode duty cycle estimates.

The RL component of QSH is described at a conceptual level. Training methodology, reward function design, and model architecture are not disclosed. The performance results presented are from a deployed RL agent operating on live hardware, not a simulated RL agent evaluated on the fleet simulator.

7.3 Implications for Heat Pump Control Standards

The present results suggest three specific improvements to UK residential heat pump control practice that are achievable within current MCS and Building Regulations frameworks:

- **Replace fixed flow with weather compensation universally.** The fleet evidence base (348,170 runs) confirms 28–48% energy savings. This is the highest-impact single change available from stock controller reconfiguration, requiring no additional hardware.
- **Calibrate WC curves from room-level data, not whole-building SAP estimates.** The system identification results demonstrate that SAP-derived values are systematically insufficient at room level. A calibration run using live sensor data over 24–72 hours would substantially improve WC curve accuracy.

- **Implement demand aggregation gating.** The shoulder mode finding demonstrates that firing the heat pump on individual zone calls in mild conditions operates the compressor below its efficiency peak. Aggregate demand gating is a software-only change to the control algorithm that preserves all comfort constraints whilst enabling shoulder mode efficiency gains.

8. Conclusions

This paper has presented a physics-based fleet simulation framework covering 348,170 parameterised runs of residential heat pump performance under UK conditions, a hybrid deterministic–adaptive control system (QSH) that outperforms best-practice weather compensation across all housing archetypes, and live validation data from a 205 m² installation that reveals a previously undocumented shoulder mode operating regime.

The principal findings are: (1) weather compensation delivers 28–48% energy savings over fixed 45°C flow across all UK housing archetypes, validating industry best-practice guidance on a large systematic evidence base; (2) QSH improves on best-practice WC by +1.8–2.7% COP at fleet prediction level from the deterministic layer alone; (3) demand-gated compressor operation in mild conditions produces a 24.9% COP uplift by operating the compressor in its efficient load band — the shoulder mode finding — a regime that WC cannot access by design; (4) live system identification reveals room-level thermal conductances 12–185× above SAP industry priors, confirming that WC curves calibrated from SAP heat loss calculations are systematically incorrect for real UK housing stock.

The reinforcement learning layer independently converged on the shoulder mode efficiency optimum from reward signals alone, without knowledge of the thermodynamic mechanism — a result that demonstrates the value of deploying adaptive control within a physics-based safety envelope.

All simulation code, archetype profiles, COP models, weather compensation curves, and result databases are released under Apache 2.0. The complete dataset is available at github.com/stuartj1-1981/Quantum-Swarm for independent verification and replication.

9. Data Availability

All data supporting the findings of this paper are available via the Zenodo repository (doi:10.5281/zenodo.18903154, CC BY 4.0) and the public GitHub repository at github.com/stuartj1-1981/Quantum-Swarm (Apache 2.0). The fleet results database (fleet.db, SQLite, ~2 GB) and live historian database (qsh_live.db) are provided as release artefacts. SHA-256 checksums are documented in Appendix A of the Correlation Technical Analysis (docs/QSH_Fleet_Correlation_Technical_Analysis.md). All data are publicly available; no data are available only upon request.

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